

Reproductive Physiology

Female & Male: Axes, Hormones, Gametogenesis, Cycle, Pregnancy & Clinical Correlates

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REPRODUCTIVE PHYSIOLOGY: SHARED PRINCIPLES

Male and female reproductive physiology are built on the same core design. In both sexes a single command axis, the hypothalamo-pituitary-gonadal axis, drives gamete production and sex hormone secretion. The hypothalamus releases gonadotropin-releasing hormone (GnRH) in pulses; the anterior pituitary responds with luteinising hormone (LH) and follicle-stimulating hormone (FSH); and the gonads respond with steroids and gametes while feeding signals back to keep the whole system in balance. Reading the two sexes side by side makes the shared logic obvious and the differences easier to remember.

CORE PRINCIPLE: one axis, two outputs

GnRH pulsatility is non-negotiable in both sexes. Discrete pulses maintain gonadotroph sensitivity; continuous GnRH paradoxically suppresses LH and FSH, which is exactly how GnRH agonist drugs are used therapeutically (prostate cancer, endometriosis, assisted reproduction). LH acts mainly on the steroid-producing cells (theca in the ovary, Leydig in the testis) and FSH acts mainly on the gamete-support cells (granulosa in the ovary, Sertoli in the testis).

Element	Female	Male
Axis	Hypothalamo-pituitary-ovarian (HPO)	Hypothalamo-pituitary-gonadal (HPG)
LH target	Theca cells (androgen substrate)	Leydig cells (testosterone)
FSH target	Granulosa cells (aromatase, oestradiol)	Sertoli cells (sperm support, ABP)
Selective FSH brake	Inhibin B / inhibin A	Inhibin B (from Sertoli cells)
Dominant steroid	Oestradiol, then progesterone	Testosterone
Gamete output	Cyclical (one oocyte per cycle)	Continuous (spermatogenesis)

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FEMALE ANATOMY & THE HPO AXIS

The female reproductive system integrates internal and external organs to prepare monthly for fertilisation, support gestation, and deliver offspring. The reproductive years run from menarche (first period) to menopause (12 consecutive months without menses).

Structure	Key points	Primary function
Ovaries (2)	Almond-shaped, ~3.5 x 2 x 1 cm; in the ovarian fossa	Oogenesis; hormone production (oestrogen, progesterone, androgens, inhibin)
Fallopian tubes (2)	~10 cm; fimbriae, ampulla, isthmus, interstitium; ciliated	Transport ovum; site of fertilisation (ampulla)
Uterus	Pear-shaped; fundus, body, cervix; usually anteverted	Implantation and fetal development; menstruation
Cervix	Inferior uterine segment; transformation zone	Controls access; mucus plug; dilates in labour
Vagina	~9 cm fibromuscular tube; acidic pH 3.5 to 4 (lactobacilli)	Birth canal; coitus; menstrual efflux

2.1 GnRH: the pulse generator

GnRH is a decapeptide from the arcuate nucleus, released into portal blood in regular pulses. Pulse frequency shapes the pituitary output: fast pulses favour LH, slow pulses favour FSH. The master pulse generator is the KNDy neuron network (kisspeptin, neurokinin B, dynorphin). Stress raises opioid tone, slows GnRH, and disrupts cycles.

Parameter	Follicular phase	Luteal phase
GnRH pulse frequency	~1 pulse / 70 to 90 min (fast)	~1 pulse / 3 to 4 h (slow)
Net gonadotropin effect	Favours LH	Favours FSH
Modulator	Kisspeptin drives; oestrogen speeds	Progesterone slows frequency

2.2 LH and FSH

Gonadotropin	Target	Key actions in the female
FSH	Granulosa cells	Recruits and grows follicles; induces aromatase (androgen to oestradiol); induces LH receptors pre-ovulation; drives inhibin B
LH	Theca cells, corpus luteum	Drives theca androgen production; surge triggers ovulation (~36 h later); luteinises granulosa; drives corpus luteum progesterone

The two-cell, two-gonadotropin model

Oestradiol synthesis needs both cell types and both gonadotropins. LH drives theca cells to make androgens (androstenedione, testosterone). Those androgens diffuse to granulosa cells, where FSH has up-regulated aromatase (CYP19) to convert them to oestradiol. Neither cell can make oestradiol alone, which is why deficiency of either FSH or LH impairs oestrogen production.

2.3 Feedback mechanisms

Signal	Effect	Phase
Oestradiol (low to moderate)	Negative feedback: lowers LH and FSH	Most of the cycle
Oestradiol (high, sustained >36 h)	Positive feedback: triggers the LH surge	Late follicular, then ovulation
Progesterone (luteal)	Negative feedback: slows GnRH pulses	Luteal phase
Inhibin B (granulosa)	Selectively suppresses FSH	Follicular phase
Inhibin A (corpus luteum)	Suppresses FSH	Luteal phase

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FEMALE HORMONES & THE MENSTRUAL CYCLE

3.1 Oestrogens

Type	Potency	Main source	Dominant phase of life
Oestradiol (E2)	Most potent	Granulosa cells; placenta	Reproductive years
Oestrone (E1)	~30% of E2	Peripheral conversion in adipose	Post-menopause
Oestriol (E3)	~10% of E2	Placenta (from fetal DHEAS)	Pregnancy

System	Oestradiol effects
Reproductive tract	Endometrial proliferation; thin watery cervical mucus at ovulation; breast ductal growth
Bone	Raises bone density (inhibits osteoclasts); maintains epiphyseal closure; loss at menopause causes osteoporosis
Cardiovascular	Cardioprotective premenopause: raises HDL, lowers LDL, increases nitric oxide
Metabolism / liver	Raises SHBG, TBG, clotting factors and angiotensinogen (VTE and BP relevance with exogenous oestrogen)

3.2 Progesterone

Progesterone is a C-21 steroid made mainly by the corpus luteum in the luteal phase and by the placenta in pregnancy. It is the hormone of the luteal phase, acting after oestrogen to prepare and maintain the uterus.

Target	Progesterone effect
Endometrium	Converts proliferative to secretory, decidualised endometrium; essential for implantation
Myometrium	Reduces contractility (quiescence maintains pregnancy)
Cervix	Thick, viscous mucus hostile to sperm; closes the os
Hypothalamus	Raises basal body temperature 0.2 to 0.5 C (ovulation marker); slows GnRH pulses

Day 21 progesterone test

A serum progesterone 7 days before the next expected period (Day 21 in a 28-day cycle) confirms ovulation. A level above 30 nmol/L confirms ovulation occurred. In irregular cycles, sample weekly until menstruation. Follicular phase is below 3 nmol/L; luteal phase 15 to 160 nmol/L.

3.3 Prolactin

Prolactin is a 199-amino acid pituitary hormone, unusual in being under tonic inhibition by dopamine. Suckling, TRH, oestrogen, stress, and dopamine-antagonist drugs raise it; dopamine agonists (bromocriptine, cabergoline) lower it. It drives milk synthesis and, by suppressing GnRH, causes lactational amenorrhoea. Hyperprolactinaemia (commonly a prolactinoma) causes oligo/amenorrhoea, galactorrhoea, infertility, and reduced libido.

3.4 The Menstrual Cycle

The cycle averages 28 days (range 21 to 35). Day 1 is the first day of bleeding. It divides into menstruation, the follicular (proliferative) phase, ovulation, and the luteal (secretory) phase.

Phase	Days	Dominant hormone	Ovarian events	Endometrium
Menstruation	1 to 5	Low E2 and progesterone	Corpus luteum degenerates; new follicles recruited	Functional layer shed (spiral artery spasm)
Follicular	6 to 13	Rising oestradiol (FSH-driven)	Dominant follicle selected; suppresses others	Proliferates to 8 to 12 mm
Ovulation	14 (+/-2)	LH surge (high E2)	Follicle ruptures; secondary oocyte released ~36 h after surge	Thin elastic mucus (spinnbarkeit), ferning
Luteal	15 to 28	Progesterone + E2	Corpus luteum forms; peaks days 19 to 23	Secretory, coiled glands; optimal for implantation

The LH surge: mechanism

As the dominant follicle grows, oestradiol rises. Once E2 stays above ~200 pg/mL for 36 to 48 hours, the pituitary switches from negative to massive positive feedback, GnRH receptors up-regulate, and an explosive LH surge follows. The surge (1) resumes meiosis I in the oocyte, (2) drives prostaglandin and protease synthesis to digest and rupture the follicle wall, and (3) luteinises granulosa cells to form the corpus luteum.

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OÖGENESIS, PREGNANCY & MENOPAUSE

4.1 Oogenesis and folliculogenesis

Stage	Key event / timing
Oögonia	Mitotic diploid germ cells in fetal life; peak 6 to 7 million at 20 weeks, then atresia
Primary oöcyte	Enters meiosis I, arrested in prophase I for decades until ovulation
Primordial follicle	~1 to 2 million at birth, ~400,000 at puberty; only 400 to 500 ovulate in a lifetime
Secondary follicle	Theca cells develop; antrum forms; granulosa make oestradiol, theca make androgens
Graafian follicle	Dominant follicle selected by FSH sensitivity; others undergo atresia
Ovulation	Meiosis I completed; secondary oöcyte arrested at metaphase II until fertilisation
Corpus luteum	Secretes progesterone; lasts 14 days unless rescued by hCG in pregnancy
Corpus albicans	Fibrotic scar if no fertilisation; progesterone falls, endometrium sheds

4.2 Fertilisation, implantation, and pregnancy

Event	Timing	Details
Fertilisation	12 to 24 h post-ovulation	In the ampulla; acrosome reaction; cortical reaction blocks polyspermy
Blastocyst	Day 5	Inner cell mass (embryo) plus outer trophoblast
Implantation	Days 6 to 10	Trophoblast invades secretory endometrium
hCG secretion	Day 8 to 10	Syncytiotrophoblast hCG rescues the corpus luteum, sustaining progesterone

Hormone	Source	Key roles in pregnancy
hCG	Syncytiotrophoblast	Rescues corpus luteum; stimulates fetal testosterone; TSH-like mild thyroid stimulation
hPL	Placenta	Insulin antagonist raising maternal glucose for the fetus; lipolysis; breast preparation
Progesterone	Corpus luteum then placenta	Myometrial quiescence; immune tolerance; cervical plug
Oöstriol	Placenta (fetal DHEAS)	Uterine and breast growth; raises oxytocin receptors at term

The luteo-placental shift

The corpus luteum supports early pregnancy with progesterone, but from about weeks 8 to 10 the placenta takes over as the dominant progesterone source. This handover is why a failing corpus luteum matters most in the first trimester.

4.3 Menopause

Menopause is the permanent end of menstruation from loss of ovarian follicular activity, confirmed after 12 months of amenorrhoea (average age around 51). The perimenopause begins 2 to 8 years earlier with irregular cycles as ovarian reserve depletes.

Parameter	Change	Consequence
Ovarian follicles	Depleted	Central change driving all others
Oestradiol	Falls ~90%	Hot flushes, vaginal atrophy, osteoporosis, cardiovascular risk
FSH	Rises markedly (often >25 IU/L)	Diagnostic; loss of inhibin B and oestradiol feedback
Inhibin B	Falls early	First laboratory sign of the perimenopausal transition

Symptoms of menopause

Vasomotor: hot flushes (80%), night sweats, palpitations, from disinhibition of the hypothalamic thermoregulatory centre. Genitourinary: vaginal dryness, dyspareunia, urinary urgency (genitourinary syndrome of menopause). Psychological: mood change, insomnia. Skeletal: rapid bone loss (3 to 5% per year early), fractures. Cardiovascular: loss of cardioprotection, higher LDL and CVD risk.

4.4 Female clinical correlates

Condition	Definition	Investigation pattern
PCOS	Commonest endocrine disorder in women; Rotterdam criteria (2 of 3)	High LH, high free testosterone, high LH:FSH, 12+ follicles on USS
Premature ovarian insufficiency	Ovarian failure before age 40	FSH >25 IU/L on two occasions, low E2
Hypothalamic amenorrhoea	Functional GnRH deficiency (low BMI, exercise, stress)	Low FSH, low LH, low E2
Hyperprolactinaemia	Prolactinoma, drugs, hypothyroidism	High prolactin, low LH and FSH, MRI pituitary

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MALE ANATOMY & THE HPG AXIS

Structure	Key anatomy	Function
Testes (2)	~20 mL; in scrotum 3 to 4 C below core; seminiferous tubules	Spermatogenesis; testosterone production
Epididymis	~6 m coiled duct behind the testis	Sperm maturation and storage
Vas deferens	~45 cm muscular duct in the spermatic cord	Sperm transport; site of vasectomy
Seminal vesicles	Paired glands behind the bladder	~60% of semen: fructose, prostaglandins, alkaline pH
Prostate	~20 g around the proximal urethra	~35% of semen: PSA (liquefaction), zinc, acid phosphatase
Bulbourethral glands	Pea-sized, in urogenital diaphragm	Alkaline pre-ejaculatory lubricating mucus

The spermatic cord and varicocele

The cord contains the vas deferens, testicular artery, pampiniform venous plexus, lymphatics, nerves, and cremaster muscle. The pampiniform plexus provides counter-current heat exchange to cool arterial blood. A varicocele (dilated plexus) impairs this cooling, raises testicular temperature, and is a common cause of male subfertility.

5.1 The HPG axis in the male

Level	Hormone	Action	Feedback
Hypothalamus	GnRH (pulsatile, 1/1 to 2 h)	Stimulates LH and FSH	Testosterone gives long-loop negative feedback
Pituitary	LH	Drives Leydig cells to make testosterone (cAMP, StAR)	Testosterone inhibits LH
Pituitary	FSH	Supports Sertoli cells; raises ABP and inhibin B	Inhibin B selectively inhibits FSH
Testis	Testosterone	Spermatogenesis; secondary sex characters; anabolism	Feeds back to hypothalamus and pituitary

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TESTOSTERONE & SPERMATOGENESIS

6.1 Synthesis and transport

Testosterone is a C-19 steroid made from cholesterol in Leydig cells between the seminiferous tubules. LH binding raises cAMP and drives the rate-limiting StAR-mediated transport of cholesterol into mitochondria. In plasma, about 60% is bound tightly to SHBG (inactive), 38% is loosely albumin-bound (bioavailable), and about 2% is free and active. SHBG rises with oestrogen and thyroid hormone and falls with androgens, obesity, and insulin.

6.2 Physiological actions and DHT

Phase / system	Action
Embryonic	Testosterone drives Wolffian ducts (epididymis, vas, seminal vesicles); DHT virilises external genitalia
Puberty	Testis growth, hair, penile growth, voice deepening, muscle and linear growth, then epiphyseal fusion, libido
Adult male	Maintains spermatogenesis (high intratesticular T), libido, muscle and bone, erythropoiesis, prostate
Bone	Direct anabolic effect plus indirect via aromatisation to oestradiol (male bone health)

Testosterone versus DHT versus oestradiol in males

Testosterone is the principal circulating androgen, acting directly through the androgen receptor in muscle, kidney, and prostate. DHT is formed from testosterone by 5-alpha-reductase, is about 3 times more potent, and mediates prostate growth, male pattern baldness, external genitalia virilisation, and body hair; 5-alpha-reductase inhibitors (finasteride, dutasteride) target it. Oestradiol, formed from testosterone by aromatase, is essential for male bone density, libido, and negative HPG feedback.

6.3 Spermatogenesis

Spermatogenesis transforms diploid spermatogonia into haploid spermatozoa in the seminiferous tubules. The full process takes about 74 days plus 12 to 21 days of epididymal maturation, and runs continuously from puberty.

Stage	Ploidy	Key event
Spermatogonia (type A to B)	Diploid	Mitotic amplification and stem-cell renewal; FSH promotes differentiation
Primary spermatocyte	Diploid (replicated)	Enters meiosis I; crossing over; longest stage
Secondary spermatocyte	Haploid	Brief; immediately enters meiosis II
Spermatid	Haploid	Non-dividing; undergoes spermiogenesis
Spermiogenesis	Haploid	Acrosome and flagellum form; nucleus condenses; cytoplasm shed
Spermiation	Haploid	Mature (still immotile) sperm released into the lumen

6.4 Sertoli cell functions

Function	Detail
Blood-testis barrier	Tight junctions split the tubule into basal and adluminal compartments; protect sperm from autoimmune attack
Nutritional support	Supply lactate, pyruvate, and growth factors to avascular germ cells
Phagocytosis	Remove residual cytoplasm and apoptotic germ cells
Hormone secretion	Inhibin B (lowers FSH), androgen-binding protein (concentrates testosterone), AMH in fetal life

6.5 Sperm biology and semen

A sperm has a head (condensed nucleus plus acrosomal cap of hydrolytic enzymes), a midpiece packed with mitochondria for ATP, and a flagellar tail. Semen is mostly seminal vesicle and prostate fluid. Capacitation in the female tract removes membrane cholesterol and switches on hyperactivated motility; contact with the zona pellucida (ZP3) triggers the acrosome reaction, releasing enzymes that digest the zona so one sperm can enter.

Semen parameter	WHO lower reference limit
Ejaculate volume	1.5 mL
Sperm concentration	16 million/mL
Total sperm number	39 million per ejaculate
Progressive motility	30%
Normal morphology (strict)	4%
pH	7.2

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MALE SEXUAL FUNCTION & CLINICAL CORRELATES

7.1 Erection and ejaculation

Erection is a vascular event driven by the parasympathetic system. Sexual stimulation (psychogenic or reflexogenic via S2 to S4) releases nitric oxide from endothelium and NANC nerves. NO raises cGMP, which relaxes cavernosal smooth muscle; sinusoids fill with blood and compress venous outflow, producing rigidity. Detumescence follows sympathetic noradrenaline acting on alpha-1 receptors, and PDE5 degrades cGMP, which is why PDE5 inhibitors (sildenafil, tadalafil) treat erectile dysfunction.

Phase	Neural control	Event
Emission	Sympathetic (T10 to L2)	Semen propelled into the prostatic urethra; bladder neck closes
Ejaculation	Somatic (S2 to S4, pudendal)	Rhythmic bulbocavernosus contractions expel semen
Orgasm	CNS	Cortical sensation timed with ejaculation

7.2 Male hypogonadism

Type	LH/FSH	Testosterone	Causes
Primary (hypergonadotrophic)	HIGH	LOW	Klinefelter (47,XXY), cryptorchidism, mumps orchitis, chemo/radiotherapy
Secondary (hypogonadotrophic)	LOW or inappropriately normal	LOW	Kallmann syndrome, hyperprolactinaemia, anabolic steroid abuse, opiates
Late-onset	Mildly high	Low or low-normal	Ageing, obesity (adipose aromatase), type 2 diabetes

Features of testosterone deficiency in adult males

Sexual: low libido, erectile dysfunction, reduced ejaculate, infertility. Physical: lower muscle mass, more central fat, reduced body hair, gynaecomastia. Bone: osteoporosis (via reduced oestradiol). Mood: fatigue, depression, poor concentration. Blood: mild normocytic anaemia.

7.3 Vasectomy

Vasectomy divides and ligates the vas deferens bilaterally, preventing sperm reaching the ejaculate (failure rate below 0.1%). It does not affect LH, FSH, or testosterone, because Leydig cell function and the HPG axis are intact, and semen volume barely changes since most fluid comes from the seminal vesicles and prostate. It is effective only once semen analysis confirms azoospermia, usually after about 3 months. Reversal success falls with time, from around 90% within 3 years to about 50% beyond 9 years.



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Work through one section at a time. Read the prose for understanding, then use the tables and callout boxes for rapid revision and last-minute recall. Coloured boxes flag core principles, clinical pearls, drug targets, and emergencies. Every fact here has been rewritten in original wording from primary physiology teaching material.